



QUALITY INTELLIGENCE

Arad Quality Intelligence

Setup & User Guide

Install, configure, and run your platform — from start to finish.

Written for quality engineers. No coding required.

VERSION	Version 1.0 · June 2026
AUDIENCE	Quality Engineers & IT / whoever installs it
COVERS	Install → Setup → Connect tools → Data → Testing

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TIP New here? Read in order. If the platform is already installed and you can open it in a browser, jump to **Part 2**.

1 What is Arad Quality?

Arad Quality Intelligence is an **autonomous quality-control platform**. It continuously watches your measurement data, runs Statistical Process Control (SPC) and Gage R&R (GR&R) analysis, raises an alert the moment a process drifts out of control, and explains what's happening in plain language with an AI copilot. You drive everything from one web dashboard.

Who does what

The person who installs it (you or IT)	Puts the platform on a server or PC, sets a few security passwords, and starts it.	Once, at the start
You — the quality engineer	Sign in, connect your alert tools, bring in your data, set up gages and processes, run studies.	Day to day, in the browser

TIP You do **not** need to be a programmer. Once it's installed, everything in this guide is done by clicking around the dashboard.

2 Before you start — what you'll need

Gather these before the first-time setup. Most are optional and can be added later.

- A computer or server to run it on, with **Docker Desktop** installed — this is the only technical prerequisite.
- About **30 minutes** for the one-time install.
- Your measurement data as a **CSV or Excel file** (or your machine / MES system's API details if IT will connect it directly).
- *Optional* — sign-in details for any tools you want alerts sent to: a **Slack** webhook link, your **email** (SMTP) details, a **Twilio** account for texts, and a free **Google Gemini** key for AI summaries.

NOTE Don't have the optional pieces yet? That's fine — install first, then add each tool later whenever you're ready. Nothing breaks if you skip them.

3 Part 1 · Install the app (one-time, technical)

This part is done once by whoever sets up the server. If your platform is already running and you can open it in a browser, skip to **Part 2**.

1. Install **Docker Desktop** (Docker Engine 25+, Compose 2.20+) on the host machine. Make sure ports **3000**, **8000**, **5432** and **9092** are free.
2. Copy the project folder onto the machine and open a terminal inside it.
3. Create your settings file from the template:

```
cp .env.example .env # Windows PowerShell: Copy-Item .env.example .env
```

4. Open **.env** and set the security values below (at minimum the first four). Tip: set **SEED_DEMO_DATA=true** for the very first run so you have sample data to explore.

API_AUTH_KEY	A long random secret (32+ characters)	Lets the dashboard talk to the server securely
JWT_SECRET	A different long random secret	Protects user logins
POSTGRES_PASSWORD	A strong database password	Protects your stored data
ADMIN_USERNAME / ADMIN_PASSWORD	Your first sign-in details	Creates the first dashboard login automatically
SEED_DEMO_DATA	true (first run) / false	Loads sample measurements so you can explore

TIP Need a strong random secret? Run: `python -c "import secrets; print(secrets.token_urlsafe(48))"`

5. Start everything with one command (the first run downloads and builds the services — a few minutes):

```
docker compose up --build -d
```

6. Wait until it reports ready, then open the dashboard at **http://localhost:3000**.

IMPORTANT Keep your **.env** file private — it holds passwords. Never email it or commit it to a public place.

Useful addresses

http://localhost:3000	The dashboard — where you do your work
http://localhost:8000/health/ready	Server health check (should say "ready")
http://localhost:3002	Grafana — optional system monitoring for IT

Running in the background: the database (TimescaleDB, 5432), the data stream (Kafka, 9092) and the cache (Redis, 6379). You don't interact with these directly.

4 Part 2 · First sign-in & the welcome tour

1. Open **http://localhost:3000** and sign in with the **ADMIN_USERNAME / ADMIN_PASSWORD** you set in **.env**.

- 2. A **Getting Started** guide pops up the first time — four short steps that show you around. You can reopen it anytime from **“Getting started”** at the bottom-left of the sidebar.
- 3. Click the **Settings** gear (bottom-left). You should see a green **“Connected to your quality server.”** There's nothing to set up here — the technical address and key live under *“Advanced — for IT”* and you can ignore them.

NOTE If Settings says **“Can’t reach your quality server,”** the backend isn’t running — ask whoever installed it to check Part 1 (that Docker is up). You never fix this inside the app.

5 Part 3 · Connect your tools (fill in your API keys)

Open **Connections** from the left sidebar (under “Setup”). This is where you choose **where alerts go** and **how data comes in**. Everything you type here is encrypted on your own server and never shown again after saving. Fill in only what you use.

“Where should alerts go?” — the ones you'll most likely use

Slack	Slack webhook link	Slack → Apps → Incoming Webhooks → add to a channel
Email	Mail server (SMTP) host, port, username, password, send-from & send-to addresses	Your email / IT provider
Text message	Twilio API URL, auth token, from & to numbers	Your Twilio account

“Advanced connections” — usually set up once by IT

JIRA	JIRA address, account email, API token, project key	Auto-create a ticket when a study fails or a problem repeats
QMS	QMS events URL	Send quality events to your quality system
Automatic data feed (MES)	Measurements API URL, token, field map	Pull measurements automatically (see Part 4)
AI summaries (Gemini)	Gemini API key	Plain-English write-ups on results and alerts

To connect any tool

- 1. Type the values into that tool's box.
- 2. Click **Save changes** (top right).
- 3. Click **Send test** on that tool. A green **“Working”** badge means it's set up correctly. **“Test failed”** means re-check the value and try again.

TIP For a free AI key, open **Google AI Studio**, create an API key, and paste it into “AI summaries.” SPC and GR&R work fine without it — you just won't get the AI narratives.

6 Part 4 · Get your data in

The platform analyses whatever measurements you give it. There are three ways to get data in — pick whichever fits.

A. Upload a spreadsheet (easiest)

On the **Connections** page, under “Upload a spreadsheet,” click **Choose file** and pick a CSV or Excel export. Columns are detected automatically and the monitor starts analysing the new data right away. One row per measurement:

timestamp	2026-06-16 08:30	Yes
part_number	PN-1042	Yes
measured_value	12.503	Yes
characteristic_name	Bore diameter	Recommended
nominal_value	12.500	Optional
unit	mm	Optional
equipment_id	CNC-LATHE-07	Optional
operator_id	OP-3	Optional

TIP Column names don't have to match exactly — the app auto-detects common variations. Duplicate rows are skipped, so re-uploading the same file is safe.

B. Automatic feed from your MES / QMS (hands-off)

IT enters your system's API details under “**Automatic data feed**” on the Connections page. The app then pulls new measurements on a schedule — no manual uploads.

C. Live stream / push (advanced)

For real-time machine feeds, your systems can stream measurements to the platform's data channel or push them to its API. Your IT team can set this up using the project's technical docs.

NOTE However the data arrives, it lands in one place and the **autonomous monitor analyses it automatically** — running SPC checks and raising alerts without anyone clicking “analyse.”

7 Part 5 · Set up your quality workspace

With data flowing, set up the things that match your shop floor. All of this is in the left sidebar.

- **Gage Registry** (Quality Tools): add your inspection gauges and fixtures with calibration status. Launch a GR&R study for any gage in one click.
- **GR&R Studies** (Quality Tools): a guided three-step wizard. Type in measurements, paste a block straight from Excel, or click **Load sample** to try it. You get pass/fail results plus an AI review.
- **SPC Monitor** (Monitor): live control charts per process. Standard rules (like the Nelson rules) flag out-of-control points; click a flagged point to see the rule, the recommended action, and to acknowledge it.
- **Alert Rules** (Setup): decide which alerts reach which people or channel — e.g. send critical alerts to the night-shift Slack and email the rest. You connect the channels themselves in Part 3.

TIP Start small — add a couple of gages and one or two processes, confirm alerts reach you, then expand.

8 Part 6 · Test that everything works

Run through this checklist to confirm the whole system works end-to-end. Tick each one.

1	Server connected	Settings (gear)	Green “Connected to your quality server”
2	You can sign in	Open dashboard, log in	Dashboard loads (your initials top-right)
3	Data is in	Upload sample file on Connections	“Added X of Y rows” confirmation
4	SPC is live	Open SPC Monitor	A control chart appears for your process
5	Alerts reach you	Connect a channel → Send test	Green “Working” badge + you receive it
6	A study runs	GR&R Studies → Load sample → Run	Results + AI summary appear
7	Alerts show up	Open Alert Inbox	Active alerts listed (and counted on the bell)
8	Everything is logged	Open Audit Trail	Recent actions appear with time + actor

TIP If all eight pass, the platform is fully working and ready to hand over.

9 Troubleshooting

Settings shows “Can’t reach your quality server”	Backend isn’t running	Ask IT to confirm Docker is up; re-run docker compose up -d
“Test failed” on a channel	Wrong value (webhook, password, token)	Re-enter the value, Save changes, Send test again
Uploaded a file but see no data	Missing key columns	Ensure the file has a timestamp, a part number and a measured value
AI summaries are blank	No Gemini key	Add a Gemini key under “AI summaries” (optional feature)
Dashboard won’t open at all	Wrong address, or app still starting	Confirm <code>http://localhost:3000</code> and that the install finished

10 Glossary

SPC (Statistical Process Control)	Live charts that watch a process and flag when it drifts outside normal limits
GR&R (Gage R&R)	A study that checks whether your gauge gives consistent, trustworthy readings
Gage	An inspection instrument — caliper, micrometer, CMM, and so on
Control limits / Nelson rules	Standard statistical rules used to decide a point is “out of control”
Webhook	A web link a service gives you so another app can post messages to it (e.g. Slack)
API key / token	A secret password that lets two systems talk to each other
MES / QMS	Your factory’s Manufacturing Execution / Quality Management System
Audit trail	A tamper-proof log of every action, for traceability and audits

11 Quick reference

Open the dashboard	<code>http://localhost:3000</code>
Check the server is healthy	<code>http://localhost:8000/health/ready</code> → “ready”
Start the platform	<code>docker compose up --build -d</code>
Stop the platform	<code>docker compose down</code>
View server logs	<code>docker compose logs api</code>
Reopen the in-app tour	“Getting started” (bottom-left of the sidebar)
Confirm a tool works	Connections → Send test → “Working”

NOTE In a sentence: install once, connect your tools, bring in your data — and the platform watches quality for you, around the clock.